# Towards Clinical Application of Enhanced Timed up and Go With Markerless Motion Capture and Machine Learning for Balance and Gait Assessment

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***Abstract*—Balance and gait impairments play a key role in falls among the elderly. Traditional clinical scales such as the Berg Balance Scale (BBS) to assess fall risk are often subjective, time consuming, and do not assess gait performance. Shorter assessments such as Timed Up and Go (TUG) are available, but most clinicians only look into the completion time. This study aimed to develop a fast, low-cost, and automated framework for balance function assessment and comprehensive gait analysis by enhancing the traditional TUG test with a markerless motion capture (MoCap) system and machine learning models. In total, we included TUG datasets of 70 participants with varying degrees of fall risk based on the BBS scores. We segmented TUG trials into five phases automatically using data from the MoCap system and extracted features from the phases. These features were then analyzed to identify those that significantly discriminate between high and low fall risk groups. Using the identified features, various machine learning models were tested to estimate the BBS scores. The markers obtained from the markerless MoCap system were used for detailed gait analysis, and lower limb kinematics were compared between the markerless and marker-based methods. Our findings indicate that individuals at high risk of falling had longer completion times, lower performance velocities, and smaller ranges of motion in lower-limb joints. Among the tested machine learning models, random forest demonstrated the best performance in predicting BBS scores (RMSE: 0.98, $R^2$: 0.94). Additionally, our markerless MoCap system showed comparable accuracy to state-of-the-art systems, eliminating the need to attach markers or sensors. The findings could help develop a quick and objective tool for balance and gait assessment in older adults, providing quantitative data to improve screening and intervention planning.**

***Index Terms*—Berg balance scale, gait analysis, kinematics, machine learning, markerless motion capture system.**

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## I. Introduction

A FALL is defined as 'an unexpected loss of balance resulting in coming to rest on the floor, the ground, or other lower level' [1]. Falls in older adults are a global health and societal challenge. According to the World Health Organization, approximately one in three adults aged 65 and above experiences a fall each year [2]. Furthermore, approximately 30% of these falls result in moderate to severe injury, such as fractures and head trauma, with long-lasting and debilitating effects. Falls can lead to hospitalization, decreased mobility, reduced quality of life, and a subsequent loss of independence, incurring substantial financial burdens on the healthcare system. The causes of falls among older adults are multifaceted and complex. Among these, balance function and gait performance are critical indicators of fall risk. We posited requirements for a desirable balance and gait assessment tool, i.e., the test must be quick, accurate, and

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comprehensive, probing balance and mobility functions. The outcomes not only identify older adults at increased risk of falling but also serve as the foundation for developing early and tailored interventions to mitigate their vulnerability [3], [4].

In clinical practice, the Berg Balance Scale (BBS) and Timed Up and Go (TUG) test are widely used to assess an individual's balance function [5], [6], [7], [8], [9]. The BBS is often regarded as the "gold standard" for evaluating balance due to its comprehensive assessment of multiple dimensions of postural control, including static and dynamic balance. It consists of 14 tasks designed to evaluate an individual's ability to maintain stability during different activities. Each task is scored on a scale from 0 to 4 (where 0 indicates an inability to complete the task and 4 indicates smooth execution). The total BBS score can reach a maximum of 56, with clinical BBS scores below 50 generally indicating a high risk of falls [10], [11], [12]. Despite its widespread use and rigor, the process is time-consuming, with a full protocol taking approximately 30 minutes [8], [9]. Additionally, BBS does not directly assess gait performance. In contrast, the TUG test is a quicker and simpler alternative, measuring the time required to complete a sequence of functional movements: rising from a chair, walking, turning, and sitting back down. By incorporating elements of both balance and mobility, the TUG provides valuable insights into fall risk with a shorter time and minimal equipment. Importantly, studies have demonstrated a strong correlation between TUG and BBS scores [13], [14], [15], suggesting that the TUG test could serve as a practical proxy for predicting BBS outcomes while providing a quicker alternative. However, the reliability of the TUG can vary due to its dependence on the clinician's judgment and environmental conditions [16]. Moreover, the threshold for discriminating between low and high fall risk individuals using TUG range widely in the literature, from 10 to 25 seconds [17], [18], [19]. Given these challenges, there is a growing demand for innovative, technology-driven solutions to enhance the objectivity and efficiency of balance function assessments. The integration of automated approaches into the TUG test may enhance its utility by not only measuring completion time but also extracting various kinematic parameters from each section of the TUG movements, which have a high overlap with most components of the BBS test, such as sitting balance, sitting to stand, standing to sit, and turning, thus bridging the gap between simplicity and comprehensiveness in balance and gait evaluation.

Clinical gait analysis is commonly used to provide key insights into mobility, stability, and functional performance [20], [21], [22], [23]. Compared to younger adults, older individuals require greater motor control to maintain stability while walking, which makes gait impairments one of the strongest predictors of falls. Past studies have identified changes in specific gait parameters, such as stride length, gait speed, and variability, as independent risk factors for falls. These subtle gait changes are often imperceptible through visual observation, necessitating the use of objective and precise analytical tools [24]. While highly accurate, gait analysis has traditionally relied on marker-based 3D motion capture (MoCap) systems, which have significant limitations, such as the time-intensive process of placing retroreflective markers, the need for trained operators, and the potential interference of markers with natural movement [25]. Although wearable sensors are also widely used in balance function prediction and gait analysis [8], their placement can introduce uncertainty, causing discomfort and affecting natural movement [9]. As a result, there is increasing demand for fast, non-invasive, and user-friendly alternatives to overcome these challenges.

In recent years, markerless MoCap systems have transformed motion analysis, offering precise assessments of human movement without the need for physical markers or sensors [25]. Among these, OpenPose can detect 25 2D body keypoints for basic movement analysis, but cannot directly determine body segment orientations, has limited precision for distal joints, and struggles with occlusion or complex environments [26], [27]. OpenCap enhances this by incorporating Long Short-Term Memory networks, trained on a large MoCap dataset, to infer and "augment" the sparse keypoints with 43 detailed anatomical markers. However, because these markers are inferred rather than directly extracted from the user's images, they may introduce biases toward average movement patterns from the training data. This limits the system's generalizability, particularly for individuals with unique or pathological movement profiles [28]. Another notable system, Theia3D, predicts 3D joint positions from users' video data by using a customized dataset of annotated anatomical landmarks and scaling an articulated multibody model to match subject-specific landmarks in 3D space [29]. However, it relies on proprietary landmarks and outputs only joint centers, limiting the accuracy of landmark detection and the analysis of full-body segment orientations. In contrast, our recent work integrates anatomical priors and data-driven methods to capture human movement from multi-view RGB videos [30]. This markerless system is trained using the RRIS40 dataset, which uses 2D annotations of 40 surface anatomical landmarks projected from marker-based motion capture at pixel-level accuracy. With a multi-view weighted triangulation, this system predicts 3D marker positions with an average error of 13.23mm. Unlike Theia3D, our system directly outputs virtual marker positions corresponding to bone landmarks, which facilitates the application of well-established biomechanical formulas developed over decades. This unique capability enables comprehensive parameter derivation from movement data, enhancing biomechanical analysis. Therefore, a question arises whether an enhanced TUG test, integrating a markerless MoCap system and machine learning, could effectively predict the BBS score with high accuracy while concurrently providing a comprehensive analysis of gait and balance.

Our objectives are twofold: (1) to evaluate the agreement of lower limb joint kinematics during the walking phase of the TUG using the markerless MoCap system, compared to a marker-based MoCap system; (2) to develop a quantitative balance assessment framework for older adults by implementing an enhanced TUG test integrating a markerless MoCap system and machine learning techniques. This framework aimed to estimate BBS scores and simultaneously offer comprehensive gait analysis, including spatiotemporal parameters and joint kinematics.

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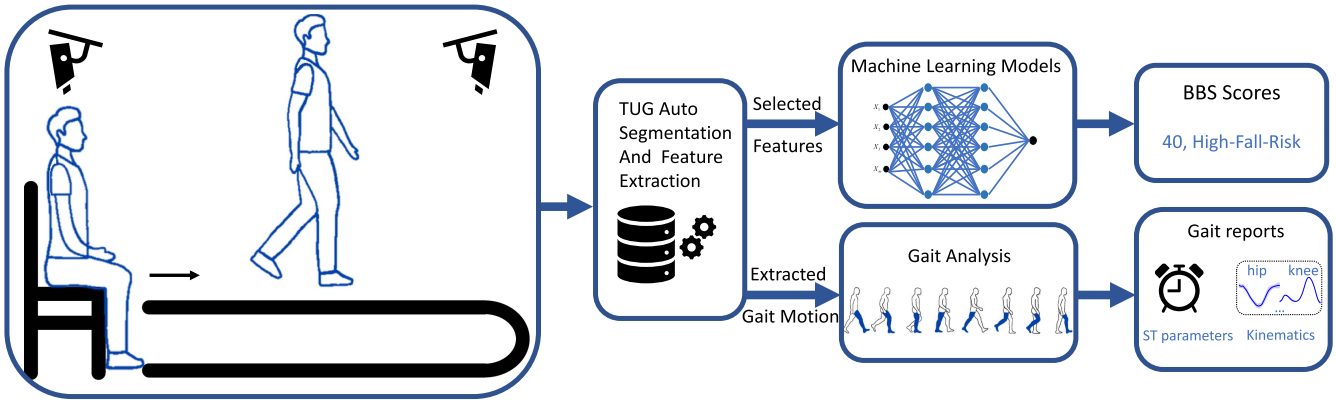



Fig. 1. Schematic of the balance and gait assessment framework using an enhanced TUG test with a markerless motion capture system and machine learning. This system can predict BBS scores and simultaneously offer comprehensive gait analysis, including spatiotemporal (ST) parameters and joint kinematics.

## II. Methods

We developed a quantitative framework for balance function assessment and gait analysis using an enhanced TUG test, integrating a markerless MoCap system and machine learning techniques (Fig. 1). TUG motions were recorded with a 3D markerless MoCap system. Using the TUG auto-segmentation approach, we obtained selected features and motion trajectories during the TUG test. These features were then used as inputs for machine learning models to predict BBS outcome, which is more widely accepted as the 'gold standard'. Walking motions were extracted from the motion trajectories, and gait analysis, including spatiotemporal parameters and joint kinematics, was conducted. The study results consist of three parts: (1) TUG features extracted from high and low fall-risk groups; (2) BBS prediction performance using commonly used regression machine learning models; and (3) a pilot study comparing lower limb joint kinematics captured by marker-based and markerless MoCap systems.

### A. Experimental Setup

In this multicentre study, we included 70 participants comprising three parts: 44 able-bodied subjects (age: 45.8 ± 20.0 years; sex: 26F/18M; weight: 60.6 ± 12.5 kg; height: 163.7 ± 10.0 cm) from the Rehabilitation Research Institute of Singapore (approved by the Nanyang Technological University Institutional Review Board, IRB-2022-382), 7 post-stroke individuals (age: 65.6 ± 5.6 years; sex: 3F/4M; weight: 57.9 ± 5.1 kg; height: 167.4 ± 4.1 cm) from Hong Kong Polytechnic University (approved by Human Subjects Ethics Subcommittee of Hong Kong Polytechnic University, HSEARS20210715003-01), and 19 older adults with knee osteoarthritis (age: 62.3 ± 9.2 years; sex: 17F/2M; weight: 57.4 ± 9.7 kg; height: 156.3 ± 10.0cm) from the Guangdong China-Singapore Intelligent Rehabilitation Institute (approved by Foshan Jianxiang Nursing Home Medical Ethics Committee, IRB-2023-02). These participants were instructed to perform TUG movements within a 3D MoCap system. All participants were able to complete the experimental procedures as directed by the instructors and willingly consented to participate in the study by signing informed consent forms. All BBS assessments were performed by a professional physical therapist.

TUG trials were recorded using the Vicon (200Hz) system in Hongkong and OptiTrack (200Hz) system in Foshan. TUG trials were recorded in Singapore simultaneously by a marker-based Miqus M3 Qualisys (200Hz) system and a markerless MoCap system [30]. The markerless MoCap system comprises 8 cameras and uses computer vision to automatically capture 3D data of human movement. The system can obtain kinematic data by automatically extracting the 2D position of each virtual anatomical marker from video captured by the multi-camera system and triangulate them. All trials were performed inside the capture volume of the markerless system.

In each trial, participants performed TUG for 3 repetitions. In each repetition, they were initially seated on a height-adjustable chair such that the knee was at a 90-degree angle. After the cue from the experimenter, participants then stood up and walked for a distance of 3 meters towards a line marked on the floor. They were asked to walk as fast as possible but safely. Upon passing the line, the participants then made a 180-degree turn to go back towards the chair. Upon reaching the chair, they would then be required to sit back, and that repetition was considered complete. The experimenter continued to cue the participants for subsequent repetitions. The repetition was repeated if there was a false start or failure to comply with the required TUG performance.

### B. TUG Segmentation

We segmented TUG into five phases: sit-to-stand, forward walking, turning, return walking, and stand-to-sit (Fig. 2(a)).

- Sit-to-stand phase: The marker placed on the C7 vertebra was used to detect the start and standing event of the sit-to-stand phase. The movement of the C7 marker in the vertical Z-direction transitions from stable to moving when the person begins to stand, which was detected as the start event. After standing up, the C7 marker stops increasing in the Z direction, which was detected as the standing event (Fig. 2(b)).

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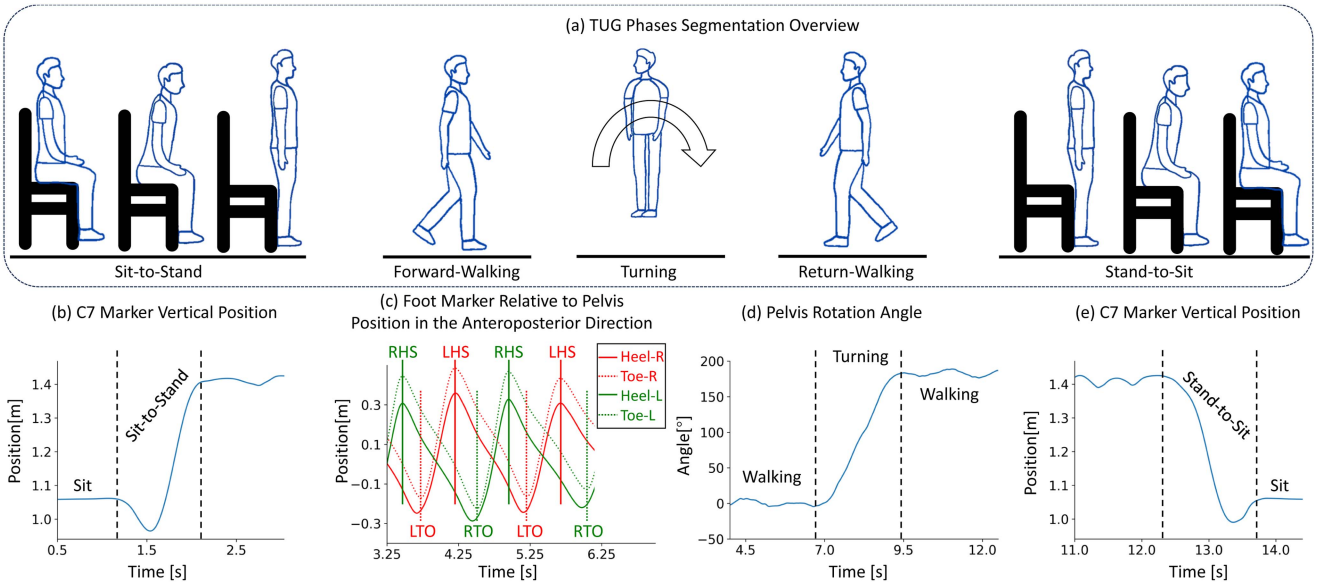



Fig. 2. Illustration of TUG phases with segmentation approaches. (a) Overview of TUG phases: sit-to-stand, forward walking, turning, return walking, and stand-to-sit. (b) Sit-to-stand phase segmentation using C7 marker's vertical Z-direction position (c) Heel strike and toe-off gait events using the heel and toe markers relative to the pelvis coordination system in the anteroposterior direction. (d) Turning phase segmentation using pelvis rotation angle. (e) Stand-to-Sit phase segmentation using C7 marker's vertical Z-direction position.

- Forward walking phase: After standing up from the chair, the subject walked 3 meters and made a turn. The heel strike (right heel strike, RHS; left heel strike, LHS) and toe-off (right toe-off, RTO; left toe-off, LTO) gait events during the gait cycle were detected using the heel and toe markers relative to the pelvis coordination system in the anteroposterior direction (Fig. 2(c)) [31].
- Turning phase: While turning, the body rotates around 180 degrees. The start and end of the turning phase were detected when the body began to rotate and complete the rotation. The pelvis rotation angle was used to segment this phase (Fig. 2(d)).
- Return walking phase: After turning back, the person walked 3 meters back to the starting point of the TUG test. The gait cycle was detected using the same method as in the forward walking phase.
- Stand-to-sit phase: Similar to the sit-to-stand phase, the standing event was detected when the C7 marker in the Z direction started to decrease, and the sitting event was detected when the Z direction of the C7 marker stabilized in its position after decreasing (Fig. 2(e)).

## C. Feature Extraction

Based on the five phases extracted from the TUG test, we selected the following parameters:

- Total duration and duration of each phase. 6 features were extracted.
- Maximum vertical velocity to stand and sit down. 2 features were extracted.
- Several features during the turn phase, including **turn steps** (the number of steps taken during the turn), **turn width** (the Euclidean distance between the pelvis positions at turn start and turn end), **turn arc length** (the length of the curved path traveled by the pelvis during the turn), and **turn speed** (turn arc length divided by the turning time). 4 features were extracted.
- Gait spatiotemporal features extracted from gait cycles, strides, and steps for each leg, including gait speed, cadence, stride length, step length, step time, step width, as well as gait cycle time, stance time, swing time, initial double support time, terminal double support time, total double support time, and their proportions relative to the gait cycle time. Additionally, the average, ratio, and gait symmetry index of these left and right leg features were calculated. 85 features were extracted. Gait features were extracted from the forward and return walking phases of the TUG test.
- Gait kinematics features extracted from lower limb joints, including the minimum, maximum, and range of motion of hip flexion/extension (F/E), knee flexion/extension, and ankle dorsiflexion/plantarflexion (D/P). Then, 18 features were extracted.

In total, 115 features were extracted from three repetitions of TUG. Gait spatiotemporal and kinematic features were averaged across gait cycles of all repetitions, while the remaining features were averaged across repetitions.

## D. Machine Learning Models/Regression Models

Linear regression (LR), least absolute shrinkage and selection operator regression (Lasso), random forest (RF), gradient boosted decision tree (GBDT), extreme gradient boosting (XGBoost), k-nearest neighbors regression (KNN), and support vector regression (SVR) were used to estimate BBS scores with selected features. These models were chosen to include both

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linear and non-linear approaches, ensuring a comprehensive evaluation of different modeling paradigms, and their selection was further motivated by their frequent application in similar studies in the literature [8], [32].

**LR** is a linear approach to modeling the relationship between a target variable and one or more independent input variables. The objective function is formulated as:

$$F_{obj} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

where $y_i$ and $\hat{y}_i$ are the actual and predicted value, and $n$ is the number of samples.

**Lasso** regression, a variant of linear regression, automatically selects features by shrinking less important coefficients to zero, making it useful when dealing with many features and selecting only the relevant ones. The objective function can be written as:

$$F_{obj} = \frac{1}{2n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \alpha\sum_{j=1}^{p}|\beta_j| \tag{2}$$

where $p$ is the number of features, $\beta_j$ is the coefficient of feature $j$, and $\alpha$ is the regularization parameter.

**RF** is an ensemble method that effectively models non-linear relationships between inputs and target variables by constructing multiple decision trees and combining their predictions. Each decision tree partitions the dataset into smaller, more predictable subsets based on input features, with internal nodes representing feature-based decisions and leaf nodes representing outcomes or predictions. Compared to a single decision tree, RF reduces overfitting and improves prediction accuracy through ensemble averaging. **GBDT**, on the other hand, is also an ensemble method but builds trees sequentially. Each new tree corrects errors made by previously trained trees, unlike RF, which trains each tree independently using a random sample of the data. **XGBoost** is a more regularized form of gradient boosting that enhances model generalization by incorporating advanced regularization techniques. The objective function for XGBoost can be written as:

$$F_{obj} = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{k=1}^{K}\Omega(f_k) \tag{3}$$

where $L$ is the loss function that measures the difference between the true target values $y_i$ and the predicted values $\hat{y}_i$; $\Omega$ is the regularization term that penalizes the complexity of the model; $f_k$ represents the individual decision trees in the ensemble.

**KNN** regression makes predictions by finding the K nearest data points to a given input and averaging their target values. Mathematically, the prediction $\hat{y}$ for a new data point $x$ in regression is given by:

$$\hat{y} = \frac{1}{k}\sum_{i=1}^{k} y_i \tag{4}$$

where $y_i$ represents the target value of the $i_{th}$ nearest neighbor to $x$, and $k$ is the number of neighbors to consider.

**SVR** is a variant of Support Vector Machines used for regression analysis. Its goal is to find an optimal hyperplane that best fits the data while maintaining a maximum margin (epsilon-insensitive tube) around the predicted values. The objective function is formulated as follows:

$$F_{\text{obj}} = \left(\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\zeta_i + \zeta_i^*)\right) \tag{5}$$

subject to: $y_i - w^T\phi(x_i) - b \le \epsilon + \zeta_i$, $w^T\phi(x_i) + b - y_i \le \epsilon + \zeta_i^*$, and $\zeta_i, \zeta_i^* \ge 0$. Where $w$ is the weight vector, $b$ is the bias term, $\phi(x_i)$ is the feature mapping function, $\zeta_i$ and $\zeta_i^*$ are slack variables, $\epsilon$ is the margin of tolerance, and $C$ is the penalty parameter for errors.

### E. Feature Selection

The subjects were categorized into two groups using a cutoff score of 50 [10], [11], [12]. Participants with a BBS score $> 50$ were classified as low-fall-risk individuals, while those with a score $\le 50$ were classified as high-fall-risk individuals. This classification resulted in 55 low fall-risk individuals and 15 high fall-risk individuals. The high fall-risk group included 7 older post-stroke individuals and 8 older individuals with knee osteoarthritis.

To assess the normality of TUG features, Shapiro-Wilk tests were used. Subsequently, two-sample t-tests and Mann-Whitney U tests were used to compare features between the high- and low-fall-risk groups for features with normal and non-normal distributions, respectively. Statistical results are considered significant for $p < 0.05$. To optimize the feature set, redundant features were identified through correlation analysis. Specifically, pairs of features with a correlation coefficient exceeding 90% were considered redundant. Within each pair, the feature demonstrating the most significant statistical distinction between the high- and low-fall-risk groups (i.e., the lowest p-value) was retained to ensure the selection of features most relevant to the analysis [32], [33].

### F. Evaluation Protocol

We evaluated the BBS score prediction accuracy of these machine learning models: LR, Lasso, RF, GBDT, XGBoost, KNN, and SVR. The data was split into 80% for training and 20% for testing. To mitigate the risk of overfitting, five-fold cross-validation was conducted, which is a widely used method to ensure a thorough assessment of the model's generalizability [34]. Root mean square error (RMSE), absolute mean error (MAE), and R-squared ($R^2$) were used to evaluate the estimation performance of each model. A low RMSE and MAE indicate high prediction accuracy. $R^2$ returns a value between 0 and 1, providing a measure of how much of the variability in the actual BBS scores is explained by the model's predictions. A value of 0 indicates no explanatory power, while a value of 1 indicates that the model perfectly explains all of the variability in the actual BBS scores.

Simultaneous markerless and marker-based data were collected for analysis from eighteen subjects during walking, extracted from TUG trials. The RMSE and intraclass correlation

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coefficients (ICC) of hip, knee, and ankle joint kinematics, calculated using both marker-based and markerless MoCap systems, were analyzed.

## III. Results

### A. TUG Features

We extracted 115 features from each TUG trial. Overall, significant differences were found in most features between the high- and low-fall-risk groups, including those related to the total duration and duration of each phase, maximum vertical velocity, turn phase, gait spatiotemporal parameters, and gait kinematics (Fig. 3). However, no significant differences were observed in the gait symmetry ratio and step width.

In terms of time duration, the high-fall-risk group exhibited longer total TUG times, longer duration in each phase, and longer times in each gait phase (all $p < 0.01$, Fig. 3(a), (b)). Regarding the fraction of gait cycle time, the high-fall-risk group had higher values for stance time, initial double (ID) support time, terminal double (TD) support time, and total double support time fractions (all $p < 0.01$, Fig. 3(c)).

In terms of velocity/speed performance, the high-fall-risk group had lower velocities/speeds for sit-to-stand, turning, gait, and stand-to-sit phases, as well as lower gait cadence (all $p < 0.01$, Fig. 3(d), (e)). Additionally, the high-fall-risk group took more steps during turning phases ($p < 0.01$, Fig. 3(f)). During gait and turning, this group also had smaller turn widths, turn arc lengths, stride lengths, and step lengths (all $p < 0.01$). However, no significant differences were observed in step width ($p = 0.37$, Fig. 3(g)).

Regarding gait symmetry features, no significant differences were observed in all gait symmetry ratios (Fig. 3(h)). However, most gait symmetry index features showed significant differences, except for initial and terminal double support times and their fractions, as well as stride length (Fig. 3(i)).

Among gait kinematics metrics, the high-fall-risk group had significantly lower ranges of motion and higher minimum values in hip F/E, knee F/E, and ankle D/P (all $p < 0.01$). No significant differences in maximum hip F/E were observed (Left: $p = 0.82$, Right: $p = 0.38$, Fig. 3(j)).

After removing redundant features, we retained 28 optimal features, including sit-to-stand time, turn time, stand-to-sit time, sit-to-stand maximal vertical velocity, turn steps, turn width, turn arc length, left and right stance time fraction, left and right swing time fraction, right initial double support time fraction, right terminal double support time fraction, right gait speed, average terminal double support time, double support time asymmetry index, gait speed asymmetry index, cadence, cadence asymmetry index, step width asymmetry index, right hip F/E range of motion, left and right knee F/E range of motion and minimum value, left ankle D/P range of motion and minimum value, and right ankle D/P maximum value. These features served as inputs for the regression models developed to predict BBS scores.

### B. BBS Score Prediction

Overall, all regression model has high prediction accuracy (RMSE $\leq 1.62$, MAE $\leq 1.38$, $R^2 \geq 0.82$, Fig. 4). Among the

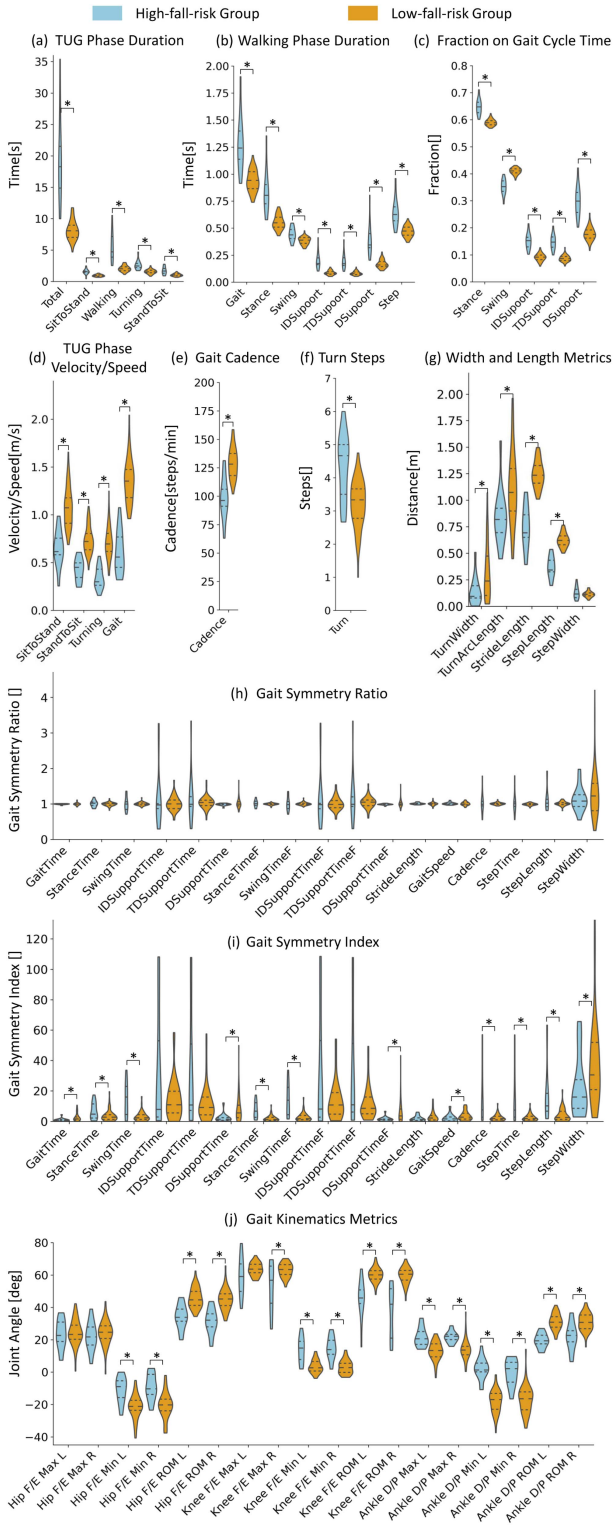



Fig. 3. Distributions of features between the high and low fall-risk groups. The features include (a) TUG Phase Duration, (b) Gait Phase Duration, (c) Gait Phase Time Fraction of Gait Cycle Total Time, (d) TUG Phase Velocity/Speed, (e) Gait Cadence, (f) Turn Steps, (g) Width and Length Metrics, (h) Gait Symmetry Ratio, (i) Gait Symmetry Index, and (j) Gait Kinematics Metrics (minimum, maximum, and range of motion of hip, knee, and ankle joints). $*$ indicated a significant difference between the two groups, determined using a two-sample t-test for normally distributed data or the Mann-Whitney U test for non-normally distributed data.

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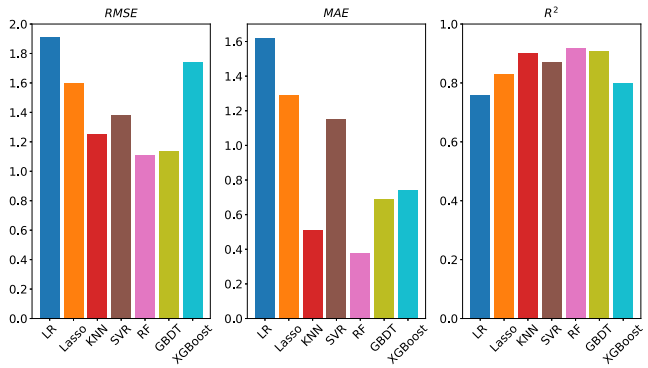



Fig. 4. Root mean square error (RMSE), absolute mean error (MAE), and R-squared ($R^2$) between predicted by machine learning models and actual BBS scores.

models, RF, GBDT, and KNN exhibit low prediction error and high explanatory power. In contrast, LR and XGboost demonstrate high prediction error and low explanatory power.

Specifically, LR was observed with the highest prediction error (both RMSE and MAE) with low explanatory power $R^2$. Lasso and SVR show similar prediction performance, with somewhat improved prediction error and higher explanatory power compared to LR.

RF demonstrates further improvement in prediction performance, with the lowest MAE among all models (RMSE: 0.98, MAE: 0.38, $R^2$: 0.94). GBDT shows similar prediction performance to RF, while XGBoost exhibits higher prediction errors and lower explanatory power.

### C. Markerless vs. Marker-Based MoCap Kinematics

Overall, the lower limb kinematics obtained by the markerless system exhibited a similar pattern to the marker-based system (average RMSE: 5.06°, Fig. 5). Hip F/E displayed an extension offset throughout the gait cycle, while knee F/E showed a similar extension offset. Ankle D/P measured by the markerless system showed good agreement with the marker-based system (average RMSE: 3.03°, standard deviation (STD): 1.26°, ICC: 0.98), compared to hip F/E (average RMSE: 9.14°, STD: 3.55°, ICC: 0.99) and knee F/E (average RMSE: 7.72°, STD: 2.45°, ICC: 0.99).

## IV. Discussion

We proposed a quantitative framework for balance function assessment and gait analysis by enhancing the traditional TUG test with a markerless MoCap system and machine learning models, eliminating the need for sensors. This system provides more clinically relevant outcome measures than the traditional TUG test, which primarily relies on completion time. Additionally, our system predicts BBS scores and offers gait analysis, including spatiotemporal parameters and joint kinematics. Our system is more time-efficient compared to the gold standard BBS, requiring significantly less time from clinicians, with assessments typically completed within 5 minutes. It also eliminates the need for specialized clinicians, making it more accessible and cost-effective for widespread use. We found that individuals at high

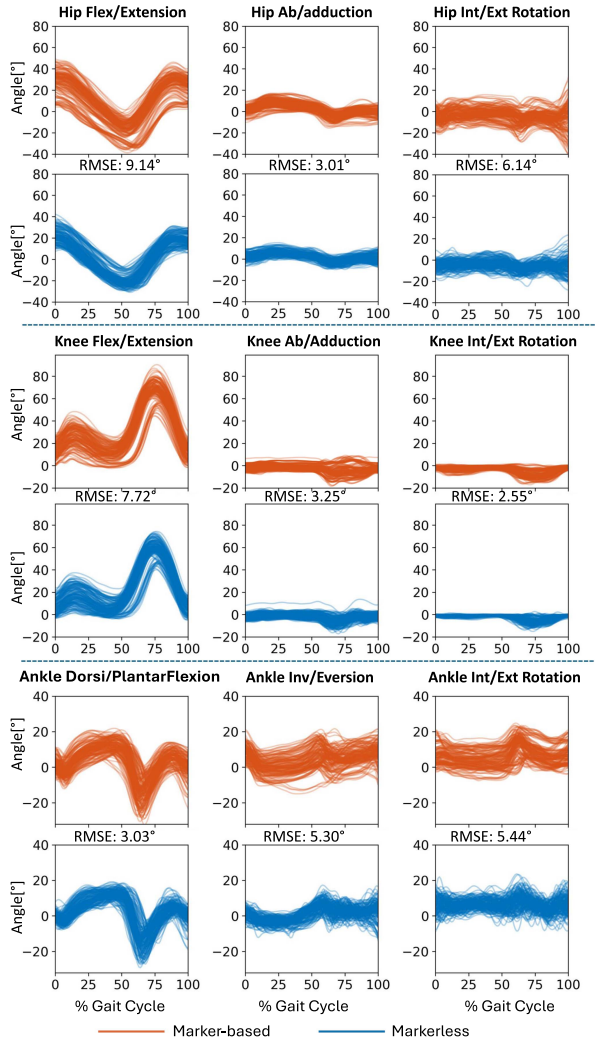



Fig. 5. Lower limb joint kinematics across 163 gait cycles measured by the markerless and marker-based motion capture system from 18 participants. Average RMSE values over gait cycles are inset in each joint angle plot.

risk of falling had longer completion times, slower performance velocities, and reduced ranges of motion in lower-limb joints. Among the tested machine learning models, Random Forest demonstrated the highest accuracy in predicting BBS scores. Additionally, our markerless MoCap system achieved accuracy comparable to state-of-the-art systems. By leveraging this comprehensive assessment framework, therapists can develop tailored interventions that precisely target stability improvement, risk reduction, and overall mobility enhancement.

Individuals with a high fall risk exhibited longer total TUG times, longer duration in each phase, and longer times in each gait phase. This finding aligns with the study by Kim et al. [4], which used a Kinect camera-based test battery to assess fall risk in older adults. They found a significant increase in turning time but no significant difference in total completion time. They attributed this to the limited trackable range of the Kinect, which resulted in a shorter TUG walking distance (2 meters) compared to the original 3 meters, thereby reducing its predictive power. Additionally, the discontinuation of the Kinect V2 camera by Microsoft in October 2017 may impact the reproducibility of

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their assessment system in other settings. In contrast, our system provides more comprehensive information, including BBS scores and gait analysis. We also observed shorter stride length, step length, cadence, and gait speed as well as longer gait time and step time during gait, which aligns with the findings of Dubois et al. [17], who reported significantly shorter step length, step time, cadence, and gait speed. They did not find significant differences in turn steps and turn width, which may be related to participants' turning habits. In our study, we did not find significant differences in step width, likely due to participants' walking habits during the TUG test where turning is involved. Compared to their study, our approach provides additional information, including BBS scores and detailed gait analysis that encompasses lower limb joint kinematics. In the high-fall risk group, most gait symmetry index features showed significant differences. However, no significant differences were observed in all gait symmetry ratios. This may be because gait symmetry ratios are not as sensitive as gait symmetry indices.

Machine learning models can provide high accuracy in predicting BBS scores. Even a linear regression model using features extracted from TUG trials demonstrated good prediction accuracy (RMSE: 1.62, MAE: 1.38, $R^2$: 0.82). Non-linear models, such as RF, showed excellent prediction accuracy (MAE: 0.38). This aligns with the findings of Lin et al. [8], who applied various machine learning models, including bagging regression, KNN, Lasso, LR, RF, ridge regression, and SVR, to predict BBS scores using features from IMU sensors. They also found the RF model had the lowest MAE (MAE: 1.19). Similarly, Badura et al. [35], using five IMUs, achieved an MAE of 1.47, suggesting that our model's accuracy meets the expected standards. In our study, RF and GBDT exhibited similar prediction performance, as both are ensemble methods capable of modeling non-linear relationships. XGBoost, however, had a slightly worse prediction performance. Its hyperparameters, including the number of boosting rounds, learning rate, maximum depth of trees, and regularization terms, were tuned using grid search. The model's suboptimal performance is probably due to the limited size of the dataset, which may have constrained its ability to fully learn from the data.

Our system demonstrated comparable accuracy to state-of-the-art technologies, with lower limb kinematics exhibiting similar patterns and ranges of motion to marker-based systems. The average RMSE was less than 9.14°, comparable to Theia3D. In a recent study investigating gait kinematics in 30 able-bodied adults using both marker-based and Theia markerless MoCap systems [36], with an average RMSE less than 13.2°, individual joint RMSE values are as follows: hip F/E: 11°, hip abduction/adduction (Ab/Ad): 2.6°, hip internal/external rotation (I/E R): 6.9°, knee F/E: 3.3°, knee Ab/Ad: 3.2°, knee I/E R: 13.2°, ankle D/P: 6.7°, ankle inversion/eversion (I/E): 8°, and ankle I/E R: 11.6°. Our system showed better prediction accuracy for the ankle joint (ankle D/P: 3.03°, I/E: 5.30°, and I/E R: 5.44°) but worse prediction for knee F/E: 7.72° and knee Ab/Ad: 3.25°, though it performed better in knee I/E R: 2.55°). Both systems demonstrated significant error in the hip joint (hip F/E: 9.14°, Ab/Ad: 3.01°, I/E R: 6.14°), likely due to the difficulty in predicting the position of the pelvis segment. In our system, the knee joint angle calculation also relies on the marker on the pelvis, resulting in a slightly higher RMSE. The slightly smaller standard deviation in the markerless mocap system (Fig. 5) may be attributed to the machine learning model's training to minimize prediction error. This process may inherently align outputs closer to the average patterns in the training data, thereby reducing variability. In contrast, the marker-based system is more prone to variability from marker placement errors.

In this proof-of-concept study, features were derived from both marker-based and markerless MoCap systems due to the limited number of subjects we were able to recruit in Singapore. The markerless system was trained on a marker-based dataset that followed the same markerset and marker placement protocol and included over 200 subjects performing various activities. This dataset is part of an Asian-centric movement database collected over five years [37]. A comparison of lower limb joint kinematics between the two systems showed similar patterns and ranges of motion, supporting the reliability of our approach. Based on this evidence, we expect minimal differences in predictive performance between the two systems. Nonetheless, further investigations with a larger sample size are needed to strengthen the robustness and broaden the applicability of the markerless MoCap system.

We acknowledge that falls among older adults have diverse and complex causes, requiring a multidimensional approach to evaluating fall risk factors. While our study provides balance function assessment and detailed gait analysis with a focus on modifiable physical factors, it does not encompass the entire spectrum of fall risk factors, such as cognition. Furthermore, some studies have demonstrated the limited predictive performance of the BBS and TUG tests for fall risk, and they are not recommended for use as standalone predictors despite their widespread application as clinical tools [5], [38]. In this study, our primary objective was to develop a fast, low-cost, and automated framework for balance function assessment and comprehensive gait analysis, recognizing their critical role in addressing modifiable risk factors. This framework combines the widely used TUG test with an advanced markerless MoCap technology and machine learning techniques. However, future research should consider incorporating a broader range of factors to establish a more comprehensive approach to fall risk assessment and prevention.

The gait event detection method used in this study was proposed by Zeni et al. [31], a widely adopted approach that relies on the heel and toe markers relative to the pelvis coordinate system. This method was chosen because it is well-suited for our dataset, where most participants demonstrate clear heel strike and toe-off events. We acknowledge that alternative methods, such as the velocity threshold approach proposed by Ghoussayni et al. [39], may provide additional insights. These alternatives could be explored in future studies, particularly when analyzing diverse subject groups.

## V. Conclusion

In this study, we developed an enhanced TUG Test incorporating a markerless MoCap system and a machine learning

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model. Our system is the first to predict BBS scores with high accuracy while providing comprehensive gait analysis, including spatiotemporal parameters and joint kinematics, offering more clinically relevant outcomes than the traditional TUG. This wealth of information allows clinicians to gain a deeper understanding of lower-limb pathology. In our comparison of high and low-fall-risk groups, we observed that individuals at high risk of falling had longer completion times, lower performance velocities, and smaller ranges of motion in lower-limb joints. Among the machine learning models tested, Random Forest demonstrated superior performance in predicting BBS scores. Additionally, our markerless MoCap system showed accuracy comparable to state-of-the-art systems, eliminating the need for attaching markers or sensors. By integrating markerless MoCap technology with established balance function assessments, this study contributes to the advancement of fall prevention strategies in older adults. It offers an innovative and quantitative approach to assessing balance function and conducting comprehensive gait analysis, facilitating early identification and timely interventions.

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